Physics of the Earth and Planetary Interiors, 11 (1975) 91–96
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

A NEW PRECAMBRIAN PALEOMAGNETIC POLE FOR NORTHERN EUROPE

G.S. MURTHY and E.R. DEUTSCH

Geomagnetic Research Laboratory, Department of Physics, Memorial University of Newfoundland, St. John's, Nfld. (Canada)

(Received July 18, 1975; revised and accepted September 19, 1975)

Murthy, G.S. and Deutsch, E.R., 1975. A new Precambrian paleomagnetic pole for northern Europe. Phys. Earth Planet. Inter., 11: 91-96.

A paleomagnetic study was made of the granitic rock farsundite, exposed in southern Norway. An objective was to test the contemporaneity of this body with the neighbouring Egersund anorthosite of presumed age about 900 m.y. Two of the nine sites sampled were rejected, as the magnetization was dominantly unstable. At the seven other sites, this unstable component was either absent or it could be equally well removed by AF or thermal demagnetization: after AF treatment, all samples from these sites were left with a very stable remanence, directed steeply upwards. This magnetization was probably acquired at the time of either emplacement or recrystallization of the farsundite. A magnetic test for anisotropy indicated that the stable remanence is misaligned with the ancient Earth's field direction by about 3°, apparently due to layering of the rock fabric. After correction for this anisotropy, the mean direction from the seven sites is $D = 341°$, $I = -82.2°$, $k = 142$, $\alpha = 5.0°$, corresponding to a paleomagnetic north pole at 43.3°S, 166.0°W, dp = 9.3°, dm = 9.7°, which lies on Spall's European polar wandering curve. The farsundite pole is not significantly different from a pole position based on the Egersund anorthosite, which supports the supposition that the two rock formations are cogenetic.

1. Introduction

Farsundite is a charnockitic body that is exposed over an area of some 400 km^2 near the southern tip of Norway. Middlemost (1968) has described the geology of the area. The farsundite outcrop is in the shape of an irregular oval elongated in the east–west direction (Fig. 1). The country rock appears to have undergone polyphase deformation, with the farsundite intrusion taking place in a catazonal environment. Barth and Dons (1960) and others have shown that this emplacement occurred after the main deformational phase. The rock is an inhomogeneous body containing various facies, with abundant xenoliths. Middlemost (1968) discussed the petrogenesis of these granitic rocks and favoured the idea that they are largely of the anorthosite kindred (Hödal, 1945). Other authors (e.g., Michot and Pasteels, 1969) have indicated that a cogenetic relationship exists between the farsundite and the Rogaland anorthosite province to the west. Middlemost (1968) identified three main facies: M-type or dark farsundite is similar to the charnockite series of the Eastern Ghats, India, and crops out in the town of Farsund and westward; L-type or light farsundite is exposed mainly east of Farsund, while the B-type, found at the western border of the outcrop, is an intermixture of farsundite and microgranite. M-type farsundite was probably emplaced earliest of the three facies. The country rock into which the farsundite intruded consists mainly of gneisses, migmatites and metamorphosed supracrustal rocks.

The farsundite is part of the Precambrian Fenno-scandian Shield. Most authors (e.g., Middlemost, 1968) place its probable time of emplacement or recrystallization in the "Sveconorwegian Regeneration Period", approximately 900 m.y. ago. The only applicable radiometric result is a $^{207}Pb/^{206}Pb$ date of 920 ± 20 m.y. on pegmatite, quoted in Neumann (1960). The present paleomagnetic study was undertaken with a view of testing the contemporaneity of the farsundite with the

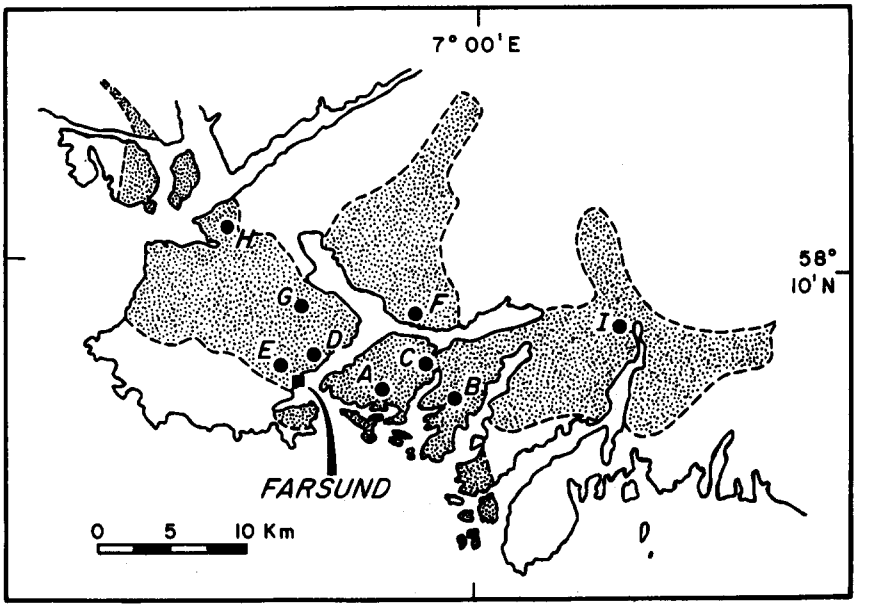


Fig. 1. Geologic sketch map of farsundite exposures (dotted areas) in southern Norway, showing paleomagnetic sampling sites $A-G$ (M-type farsundite), H (B-type) and I (L-type).

Rogaland anorthosite province. For this purpose it was necessary first to ascertain whether these rocks carry a primary component of magnetization. This was of special interest also in view of their similarity in composition with adamellite rock suites, for example those exposed near Nain, Labrador, which are currently under paleomagnetic study by us.

2. Measurements

Forty-five hand samples were collected from nine sites (Fig. 1) and oriented by solar compass. Seven sites are of M-type farsundite and one site each of B- and L-type. Two 2.5-cm diameter cylinders 2.2 cm in length were cut from each sample in the laboratory and used in the paleomagnetic measurements. The natural remanence (NRM) and low-field (0.3 Oe) volume susceptibility K_0 of all specimens were measured with a PAR spinner magnetometer and an AC susceptibility bridge. In all cases, the results for the two specimens were combined to obtain a sample mean, and each site-mean remanence direction is based on five samples.

The specimens were found to have a broad range of NRM intensities J_0 ($3 \cdot 10^{-5} - 4 \cdot 10^{-3}$ emu/cm^3) and of Koenigsberger ratios Q_0 (1–10), where $Q_0 = J_0/K_0H$ and $H = 0.50$ Oe. For six of the nine sites, the within-site NRM vectors were randomly distributed. Standard techniques of stepwise alternating-field (AF) and thermal demagnetization were applied to remove secondary components of magnetization. First, two specimens from each site were AF-demagnetized in nine steps to 850 peak Oersteds. This procedure succeeded in stabilizing the remanence directions of the M-type specimens, usually in fields $\leqslant$200 Oe (Fig. 2), but was unsuccessful in the case of the B-type and L-type farsundite. The vectors of the four specimens from these two sites moved in random fashion throughout the AF treatment. Fourteen fresh specimens (one or two per site) were then thermally demagnetized in seven steps to 600°C (Fig. 3a). In the case of the B-type and L-type sites, this treatment again failed to isolate a primary component of remanence. Such erratic response to demagnetization is typical of a very unstable magnetization, and these two sites were not used in the further analysis. However, in the case of all M-type specimens the treatment, at 400°C or less, succeeded in stabilizing the remanence directions with steep upward inclinations, closely confirm-

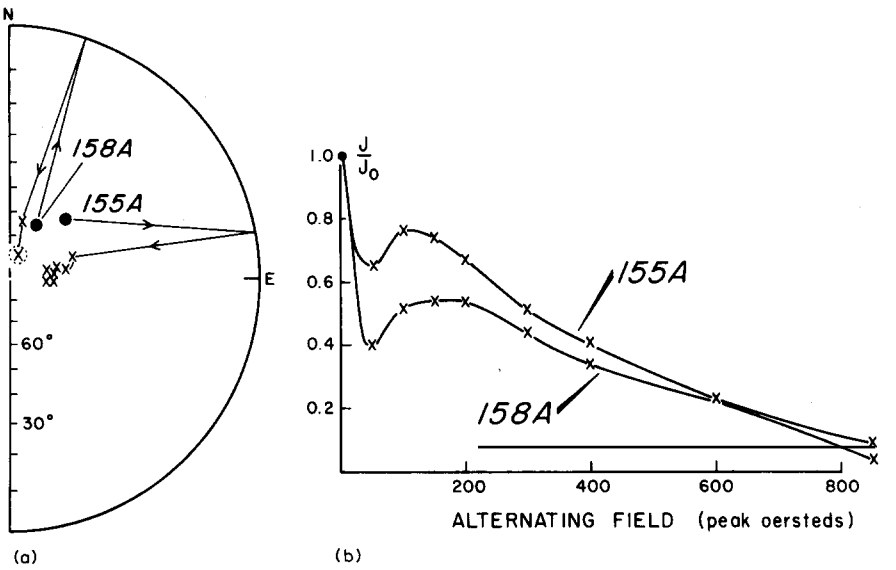


Fig. 2. Response of two farsundite specimens from site *A* to *AF* demagnetization.
(a) Change in remanence directions, plotted on a Wulff net: Dots denote positive inclinations; crosses, negative inclinations. The dashed circle indicates that essentially no directional change occurred between the 100- and 850-Oe steps.
(b) Normalized intensity curves.

ing the AF pilot results. The $J/J_{20}-T$ curves (e.g. Fig. 3a) are smooth, and the Curie balance results in Fig. 3b tend to support our assumption that the stepwise thermal treatment did not cause important physicochemical changes in the rock.

All the samples of M-type farsundite were then AF-

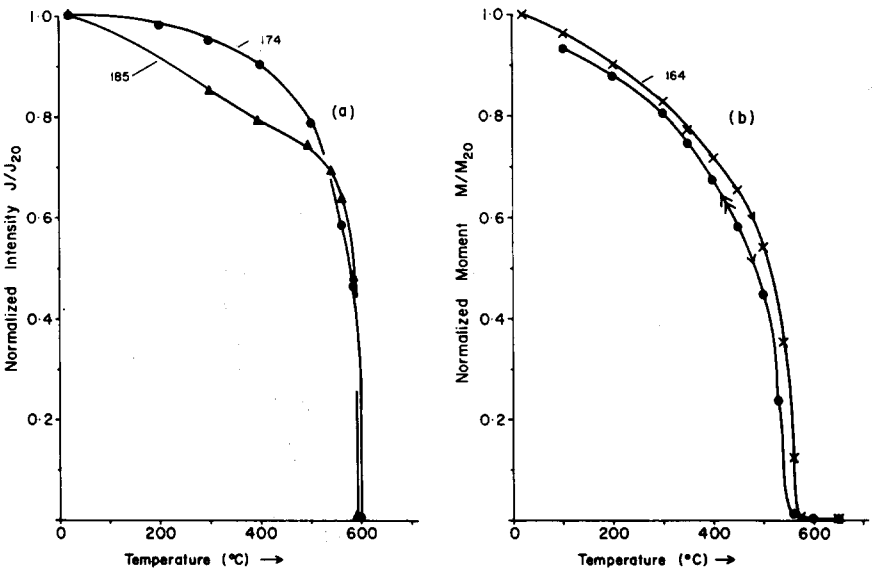


Fig. 3. (a) Stepwise thermal demagnetization in air, samples *174* (site *E*) and *185* (site *F*).
(b) Temperature dependence of magnetic moment ($M-T$ curves) measured at 1,300 Oe in air with a Curie balance. A powder of sample *164* (site *C*) was heated and cooled twice, giving $T_C = 567°$ on first heating (X's). Note that the first cooling curve (dots) and the second heating and cooling curves (same dots) are all identical.

TABLE I

Site-mean remanence directions

Site	N	NRM			$\tilde{H}$	After AF demagnetization			
		D (°)	I (°)	k		D (°)	I (°)	k	α_{95} (°)
A	5	23	−12	<2	300	73	−86	71	9
B	5	12	+50	3	200	30	−82	183	6
C	5	357	+57	3	200	32	−86	410	4
D	5	340	−63	2	200	340	−85	40	12
E	5	292	−72	14	200	254	−81	79	9
F	5	341	−77	101	250	350	−78	913	2
G	5	319	−76	61	250	320	−81	206	5
Mean, A–G	7	344.4	−46.6	2	–	348	−85.0	142	5.0
Mean, A–G *	7	–	–	–	–	341	−82.2	142	5.0

Two unstable sites have been excluded (see text). N = number of samples (two specimens per sample) or sites averaged; D = declination, east of north; I = inclination, positive down; k = precision parameter; α_{95} = radius of 95% confidence circle; $\tilde{H}$ = alternating field (AF) applied, peak Oersteds.

* Mean direction corrected for anisotropy (see text).

demagnetized, using two specimens each except for those already treated in the AF pilot study. At any site, the demagnetizing field $\tilde{H}$ was chosen to exceed by 50 or 100 Oe the value of $\tilde{H}$ at which the pilot specimens from the same site had assumed relatively stable remanence directions. Results for two specimens shown in Fig. 2 point strongly to the presence in their NRM of two oppositely polarized, though not antiparallel, components. The secondary component, which is stronger and is directed steeply downward, becomes demagnetized in fields of about 100 Oe corresponding to the peak in the J/J_0–$\tilde{H}$ curves. The steeply upward-directed component thus isolated is seen to be very stable in direction as well as intensity, the median destructive field for that component being greater than 400 Oe. Although many samples did not carry a large secondary component, the AF treatment resulted in markedly improved within-site groupings at all seven sites (Table I). The between-site directions also are well-grouped (α_{95} = 5.0°), with an overall mean direction of 348°, −85.0°.

3. Origin of the remanence

Next we measured high-field magnetic moment versus temperature (M–T) on three fresh, powdered specimens, which gave Curie points at 567, 586 and 594°C, one example being shown in Fig. 3b. This points to magnetite as the remanence carrier, though conceivably the stable part of the remanence could be due to ilmeno-hematite: Hargraves and Fish (1972) observing a similar range of Curie points in the Egersund anorthosite, attributed them to ilmeno-hematite. The M–T curves are smooth and nearly reversible, and together with all demagnetization results they indicate that the stable remanence resides in a single component. We conclude that the stable magnetization having the mean direction quoted above is probably of primary origin. In view of the advanced metamorphism of the farsundite, the stable remanence is likely to have originated at the time of its recrystallization rather than emplacement, and the term "primary" is used here to distinguish this stable component from any secondary components acquired a long time after both events.

4. Anisotropy test

Being layered, farsundite may be anisotropic and so deflect the remanence direction from the ambient field direction. We tested this with a technique described by Irving and Park (1973), using the ten samples

TABLE II

Test for anisotropy

	D (°)	I (°)	α_{95} (°)	δ (°)
Site C (N = 5):				
NRM	357	+56.8	57.8	–
200 Oe	32	−85.7	3.8	–
IRM (250, −87.7)	141	−86.7	2.4	4.6
750 Oe	186	−86.3	1.5	3.4
Site G (N = 5):				
NRM	319	−76.3	9.8	–
200 Oe	320	−80.9	5.4	–
IRM (250, −87.7)	138	−85.8	1.8	5.5
750 Oe	191	−86.6	2.4	3.0

Applied field directions are given in brackets after "IRM". N = number of samples averaged; δ = deviation of the mean direction from the applied field direction. See also text and Table I footnotes.

(one AF-demagnetized specimen per sample) from sites *C* and *G*. The specimens were placed in their relative outcrop positions between the poles of an electromagnet, where they acquired a saturation isothermal remanence (IRM) in 8,500 Oe. The field direction referred to in-situ coordinates was 250°, −87.7°, similar to the observed stable remanence directions (Table II). The IRM was then measured and the two site-mean IRM directions are seen to be very similar. However, at both sites the angle (δ) between the applied field direction and the IRM direction exceeds α_{95} of the latter, suggesting that the total remanence has suffered a small but significant deflection due to anisotropy. Since we needed to know how anisotropy may affect the stable component rather than total remanence, each specimen was then demagnetized in 750 Oe; such fields were seen earlier to be ample for isolating the stable component. Again the two site-mean directions and the saturating field direction were found to be misaligned by small but significant amounts (δ = 3.4°, 3.0°; $\delta > \alpha_{95}$). With an added uncertainty of 1° or so in the field direction this significance becomes marginal: however, the close similarity between the two 750-Oe site-mean directions suggests that the small anisotropy effect implied by the δ-values is real and that it may be similar in direction and magnitude at all seven sites. Accordingly, we assumed an anisotropy error equivalent to the 3.2° vector difference between the average value (188°, −86.4°) of the two 750-Oe site-mean directions and the saturating field direction. This difference was then subtracted vectorially from the previous overall mean direction, giving a corrected mean value of 341°, −82.2° (Table I).

5. Pole position and polarity

It seems plausible that the cooling associated with recrystallization of a large plutonic body such as charnockitic farsundite was slow enough to have averaged out the effect of secular variation between the sampling sites. We therefore justify use of the axial dipole assumption in calculating a paleomagnetic pole from the corrected mean direction. This pole is at:

43.3°S, 166.0°W, dp = 9.3°, dm = 9.7°

and is plotted in Fig. 4 along with four other Precambrian pole positions relative to Norway. The farsun-

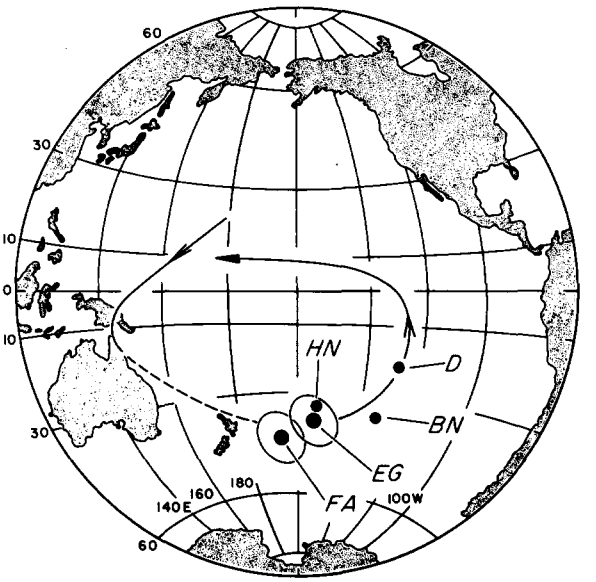


Fig. 4. Precambrian paleomagnetic poles for Norway of presumed age <850–950 m.y., with part of the polar wandering path for Europe proposed by Spall (1973). *FA*: farsundite pole, present study; *EG*: Egersund anorthosite (Murthy and Deutsch, 1974); *BN*: basement rocks of southern Norway (Poorter, 1972); *D*: Egersund dykes (Storetvedt and Gidskehaug, 1968); *HN*: Hunnadalen dykes (Poorter, 1972). Poles *FA* and *EG* are shown with 95% confidence ovals.

dite pole is not significantly different from pole *EG* (Murthy and Deutsch, 1974) based on the Egersund anorthosite, and this tends to confirm the view that the two rock formations are cogenetic. Both poles lie on the European polar wandering curve by Spall (1973), thereby adding support to the positioning of that part of his curve. Being located on that curve, the farsundite pole is most probably normal with respect to the present geomagnetic polarity.

Acknowledgements

We wish to thank Mr. W.J. Drodge (Memorial University) for assistance in all phases of the field work. We are grateful to Prof. K.M. Storetvedt (Bergen) for much helpful advice and to Dr. E. Irving (Ottawa) for reviewing the manuscript. This work was supported by National Research Council of Canada Grants A-8707 and A-1946.

References

Barth, T.F.W. and Dons, J.A., 1960. Precambrian of southern Norway. In: O. Holtedahl (Editor), The Geology of Norway. Nor. Geol. Unders., 208: 6–67.

Hargraves, R.B. and Fish, J.R., 1972. Paleomagnetism of anorthosite in southern Norway and comparison with an equivalent in Quebec. Geol. Soc. Am., Mem., 132: 141–148.

Hödal, J., 1945. Rocks of the anorthosite kindred in Vossestrand (Norway). Nor. Geol. Tidsskr., 24: 129–243.

Irving, E. and Park, J.K., 1973. Palaeomagnetism of metamorphic rocks: errors owing to intrinsic anisotropy. Geophys. J. R. Astron. Soc., 34: 489–493.

Michot, J. and Pasteels, P., 1969. The prospects of the Rb–Sr and the U–Pb methods for an advanced geochronological investigation of the Precambrian of southern Norway. Nor. Geol. Unders., 258: 17–26.

Middlemost, E., 1968. The granitic rocks of Farsund, south Norway. Nor. Geol. Tidsskr., 48: 81–99.

Murthy, G.S. and Deutsch, E.R., 1974. Rock magnetic and paleomagnetic investigations of anorthositic rocks from southern Norway. EOS (Trans. Am. Geophys. Union), 55: 234 (abstract).

Neumann, H., 1960. Apparent ages of Norwegian minerals and rocks. Nor. Geol. Tidsskr., 40: 173–191.

Poorter, R.P.E., 1972. Paleomagnetism of the Rogaland Precambrian (southwestern Norway). Phys. Earth Planet. Inter., 5: 167–176.

Spall, H., 1973. Review of Precambrian paleomagnetic data for Europe. Earth Planet. Sci. Lett., 18: 1–8.

Storetvedt, K.M. and Gidskehaug, A., 1968. Paleomagnetism and the origin of the Egersund dolerites, S. Norway. Nor. Geol. Tidsskr., 48: 121–125.